

FINAL DIAGNOSIS:

PART 1: THYROID, TOTAL THYROIDECTOMY (28 GRAMS) - $\geq 50\%$ tall cell per TSS. Collection Date:

- PAPILLARY THYROID MICROCARCINOMA, TALL CELL VARIANT (0.3 CM) CONFINED TO THE LEFT LOBE.
- METASTATIC CYSTIC PAPILLARY THYROID CARCINOMA IN LYMPH NODE, ISTHMUS; MICROSCOPIC EXTRANODAL EXTENSION PRESENT (1/1) (see comment).
- PATHOLOGIC STAGE: pT1a N1a.

PART 2: LYMPH NODES, CENTRAL COMPARTMENT, SELECTIVE NECK DISSECTION -

- METASTATIC PAPILLARY THYROID CARCINOMA IN ONE OF ONE LYMPH NODE (1/1); NO EXTRANODAL EXTENSION.
- UNREMARKABLE THYROID TISSUE.
- SPECIMEN ENTIRELY SUBMITTED FOR HISTOLOGIC EVALUATION.

COMMENT:

The isthmus nodule that was previously aspirated [redacted] with a low level BRAF mutation (V600E) is in fact the aforementioned lymph node in Part 1.

The presumed primary tumor is a 0.3 cm microcarcinoma, tall cell variant with abundant fibrosis, though not subcapsular in location. Molecular testing on this focus will be performed and reported in an addendum.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE: Total Thyroidectomy
TUMOR SITE: Left Lobe
TUMOR FOCALITY: Unifocal
TUMOR SIZE (largest nodule): Greatest Dimension: 0.3 cm
HISTOLOGIC TYPE: Papillary carcinoma, tall cell variant
PRIMARY TUMOR (pT): pT1a
REGIONAL LYMPH NODES (pN): pN1a

UUID: FF588CC0-F345-403A-AE69-97F8F3F27274
TCGA-BJ-A45J-01A-PR

Redacted



Number of regional lymph nodes examined: 2
Number of regional lymph nodes involved: 2

EXTRANODAL EXTENSION:
DISTANT METASTASIS (pM):
EXTRATHYROIDAL EXTENSION:
MARGINS:
LYMPH-VASCULAR INVASION:

Present
Not applicable
Not identified
Margins uninvolved by carcinoma
Present

ICD-O-3

carcinoma, papillary, tall cell
variant 8344/3

Sites: thyroid, NOS C73.9

Addendum

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block 11 (THYROID, LEFT LOBE, PAPILLARY THYROID MICROCARCINOMA, 0.3 CM):

- BRAF mutation IDENTIFIED (p.V600E, c.1799T>A).
- Mutations in NRAS61, HRAS61, KRAS12/13 and RET/PTC1 and RET/PTC3 rearrangements NOT identified.

NOTE:

Nucleic acids were extracted in the amount sufficient for testing.

BACKGROUND:

Mutations in either BRAF or RAS genes or RET/PTC rearrangements are found in more than 70% of papillary thyroid carcinomas (1). BRAF V600E (T1799A) mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between BRAF V600E mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the RAS genes or PAX8/PPARγ rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, BRAF V600E mutation and RET/PTC and PAX8/PPARγ rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). RAS mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

- Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. Am J Surg Pathol 2008; 30:216-222
- Xing M, et al. BRAF mutation predicts a poorer clinical prognosis for papillary thyroid cancer. J Clin Endocrinol Metab 2005; 90:6373-6379.
- Elisei R, et al. BRAF V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. J Clin Endocrinol Metab 2008; 93:3943-3949.
- Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential BRAF(V600E) mutational status in high aggressive papillary thyroid microcarcinomas. Ann Surg Oncol. 2009 Feb;18(2):240-5.
- Nikiforova MN, et al. RAS point mutations and PAX8-PPARγ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. J Clin Endocrinol Metab 2003; 88: 2318-26.
- Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV, ed. Endocrine Pathology: Differential Diagnosis and Molecular Advances. Humana Press: Totowa, 2004; 191-208.

MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the LightCycler platform to amplify BRAF codons 599-601, NRAS codon 61, HRAS codon 61, and KRAS codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. Single-step real-time RT-PCR was performed to amplify the fusion points of RET/PTC1 and RET/PTC3 rearrangements on ABI7500. If required, the mutation type was confirmed by Sanger sequencing of the PCR products on ABI3130. Amplification of the KRT7 genes was used to control quality and quantity of RNA and amount of epithelial cells present in the specimen. DNA of RNA from samples positive for each of these mutations were used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA and RNA.

Criteria	Yes	No
Diagnosis Discrepancy		
Primary Tumor Site Discrepancy		
HPAA Discrepancy		
Prior Malignancy History		
Dual/Synchronous Primary Noted		
Case is (circle):		
Reviewer Initials: BTH		
Date Reviewed: 8/30/2012		